HSLU Hochschule Luzern

Packbot

AI ROBOTICS

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Functions

The Swiss Federal Railways (SBB) has initiated the Packbot project, which aims to develop an autonomous robot capable of cutting vegetation around train tracks efficiently and safely. The project's primary goal is to ensure the robot operates without any issues or risks of ending up on the train tracks.

This project is critical for safety monitoring and vegetation management. Controlling the vegetation near train tracks is essential to prevent interference with railway operations. Currently, 5% of SBB's annual costs are allocated to managing overgrown vegetation, primarily performed manually. By automating this process, the Packbot enhances safety and significantly reduces operational costs.

The Packbot's ability to autonomously maintain vegetation ensures unity, efficiency, and safety, making it a valuable innovation for the railway industry.

Fig. 1 Packbot during Data collection

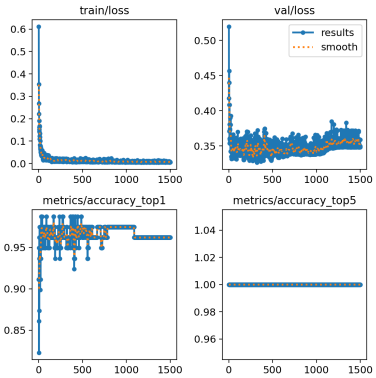


Fig. 2 Training of Yolov8n_cls

Binary Classification

Binary classification involves distinguishing between two classes, making it a focused and efficient approach for many tasks. It performs best when edge cases or challenging examples are included in the dataset, helping to improve the model's robustness. This method allows quick image processing, making it ideal for real-time applications.

Predicted

Labeled

Model Performance Across Epochs

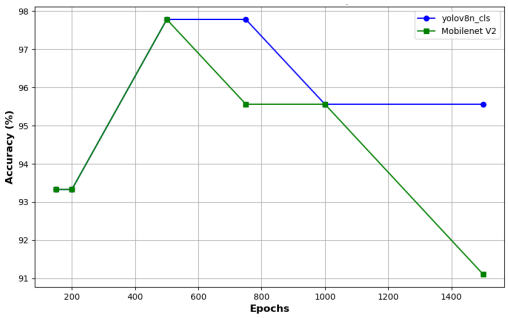


Fig. 3 Variation of accuracy depennding given the differnt Epochs for Yolov8n_cls and Mobilenet V2

Data

Data Source:

Two Rosbag recordings were conducted with the Swiss Federal Railways (SBB), complemented by an independent recording. The robot operated near and directly on train tracks, capturing diverse environments such as grass, gravel, bushes, buildings, trains, and other railway structures. These recordings provided a comprehensive dataset for training and evaluation.

Data Augmentation:

To expand the dataset and improve model robustness, several augmentation techniques were applied:

•Rotation Augmentation:

Minor rotations of ±10° were introduced to simulate real-world orientation changes without altering class labels. This helps the model generalize better to images with slight rotations.

•Mirroring:

Images were mirrored along the vertical axis to simulate flipped orientations, enriching the dataset.

•Contrast Adjustment:

Contrast levels were adjusted to make specific features either more prominent or subtle. This improves the model's ability to detect features under varied lighting conditions.

Results:

1.Expanded Dataset:

1. Training set: Increased from 760 to 2,280 images
2. Validation set: Increased from 79 to 237 images
3. Test set: Increased from 72 to 216 images

2.Improved Model Robustness:

1. The augmented dataset enabled the model to recognize images under diverse real-world conditions.
2. Enhanced the model's generalization and reliability in varied scenarios.

Model selection

Training Overview

Three distinct training phases were conducted to identify the most effective classification model:

1. Initial Screening:

Eight different classification models were trained for 150 epochs using the original dataset of 760 images. The models were then evaluated on a separate folder of generated and external images. To qualify, a model needed to achieve over 50% accuracy in binary classification ("on" or "off") on the unknown test set, ensuring it performed better than random chance (50%).

2. Refinement:

The qualifying models from the first phase were trained further for 500 epochs using the same small dataset. Four models remained:

YOLOv8n_cls (62%), **MobileNet V2** (62%), **EfficientNet B0** (51.7%), and **ResNet50** (51.7%)

EfficientNet B0 and ResNet50 showed no significant improvement with increased epochs, so the focus shifted to YOLOv8n_cls and MobileNet V2.

3. Final Selection:

The two remaining models were trained at various epochs: 150, 200, 500, 750, 1000, and 1500. The best results were observed within the range of 500 to 750 epochs.

Performance

•**YOLOv8n_cls**: Achieved **97.8% accuracy** on the validation set and **81% accuracy** on the external dataset.

•**MobileNet V2**: Achieved **97.8% accuracy** on the validation set and **81% accuracy** on the external dataset.

Epoch Analysis

•The ideal number of epochs to minimize validation loss is **500 to 750**.

•Training loss remained stable, showing no significant overtraining effect.

•Accuracy fluctuated only by **2-3%** with extended training.

Evaluation Metrics

Balanced Accuracy

$$\frac{(\text{Sensitivity} + \text{Specificity})}{2}$$

Def.: Average of recall scores for each class, giving equal weight to each class regardless of its size.

Use Case: Useful for imbalanced datasets where accuracy alone may be misleading.

Macro F1 Score

$$\frac{(\text{F1 Score of Class 1} + \text{F1 Score of Class 2})}{2}$$

Def.: Calculates the average F1 score for each class without considering class size, providing an overall F1 measure.

Use Case: Ideal for multi-class or imbalanced datasets where each class's performance matters equally.

AUPRC

Area under the Precision-Recall curve

Def.: Represents the area under the precision-recall curve, which reflects the model's ability to distinguish classes.

Use Case: Useful in imbalanced datasets to better understand the trade-off between precision and recall.

Micro F1 Score

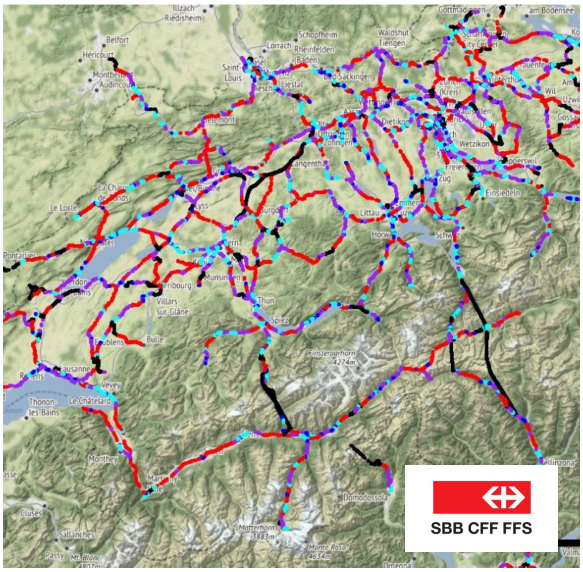
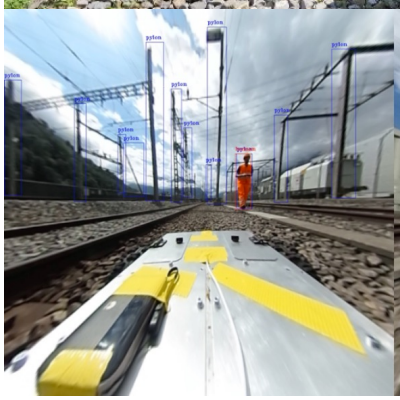
$$\frac{2*(\text{Micro Precision} * \text{Micro Recall})}{(\text{Micro Precision} + \text{Micro Recall})}$$

Def.: F1 score calculated by aggregating the contributions of all classes to compute a global measure of precision/recall.

Use Case: To have an evaluation of where both classes are equal important with an imbalanced dataset.

Vegetation Control – Weeding

Two Use Cases:

1. ***Human-Robot Collaboration:*** The robot assists by following a person walking along train tracks with a QR, following the human instructed route.

2. ***Autonomous Maintenance*:** The robot independently performs maintenance tasks and is equipped with a detection system to avoid and navigate safely away from the tracks.